FIN 550: Final Project

EXECUTIVE SUMMARY

Your Team Name (be creative):_____**Allen's Property Appraisal**_____

Select whether this is an individual or group submission. **No more than 3 members per group.** Beyond the fact that all group members may submit the same answers, each submission must be separate work.

☐ **Individual Submission**
☐ Group Submission. Group member names: _____

Case Overview

The Cook County Assessor's Office has to estimate fair market values for residential properties to determine property tax obligations. Accurate assessments are essential—overestimates burden homeowners with excessive taxes, while underestimates reduce public revenue.

The project objective is to predict the market value of 10,000 residential properties based on their characteristics. Using 50,000 historical sales records as training data, I built a model that minimizes Root Mean Squared Error (RMSE), which measures the average difference between predicted and actual sale prices.

Methodology

The training dataset contains 50,000 properties with 63 variables. I addressed missing values by replacing them with median values for numeric fields and creating a "Missing" category for categorical fields. This preserves information while allowing all records to be used in modeling.

Two key challenges shaped my approach:

First, the neighborhood variable contains 845 unique codes—too many for direct inclusion. I converted each neighborhood code to its average historical sale price (target encoding). This reduces 845 categories to a single numeric variable that captures location value. I applied the same approach to the 476 school district codes.

Second, property value depends on interactions between features. I created interaction terms (building size × bathrooms, building size × rooms) to capture these relationships. My final model uses 15 variables: building and land assessor estimates, total assessed value, building size, bathrooms, rooms, age, basement size, median income, tax rate, property type, and the engineered features above.

I compared two approaches:

1) LASSO regression produced an RMSE of $304,000. While interpretable, linear models assume straight-line relationships between features and price, which does not match housing markets.

2) Random Forest produced an RMSE of $126,000—a 58% improvement. This method builds hundreds of decision trees, each learning different patterns, then averages their predictions. It captures non-linear relationships and interactions automatically, which explains its stronger performance.

I selected Random Forest with 200 trees as my final model. Training completes in approximately 14 minutes.

Conclusion

My final Random Forest model achieves an RMSE of $126,378 on training data. Accuracy varies by price segment: for properties under $600,000 (90% of the data), RMSE is $72,088. For luxury properties over $1.5 million (1% of the data), RMSE rises to $841,525.

I recommend adopting this model for standard residential assessments while flagging properties predicted above $1 million for additional review. For future improvements, collecting additional property characteristics may help reduce errors on high-value properties.

Assessed Value Distribution

Statistic	Value
Minimum	$12,465
1st Quartile	$151,763
Median	$242,131
Mean	$317,746
3rd Quartile	$374,372
Maximum	$5,010,809

The distribution is right-skewed: most properties are valued between $150,000 and $375,000, with a small number of high-value properties extending the upper tail.

The output file "assessed_value.csv" contains 10,000 records with two columns: property ID (pid) and predicted market value (assessed_value). All values are non-negative with no missing entries.

Appendix

Table 1: RMSE by Price Segment

Prediction error increases with property value. Properties under $600K (90% of data) have RMSE of $72,088, while properties over $1.5M (1% of data) have RMSE of $841,525.

price_segment <fctr>	count <int>	pct <dbl>	rmse <dbl>
< $600K	45079	90.2	72088
$600K - $800K	2246	4.5	162333
$800K - $1M	1133	2.3	211582
$1M - $1.5M	1031	2.1	299584
> $1.5M	511	1.0	841525

5 rows

Figure 2: Distribution of Assessed Property Values

Right-skewed distribution with most properties valued between $150,000 and $400,000. A small number of high-value properties extend the upper tail to $5 million.

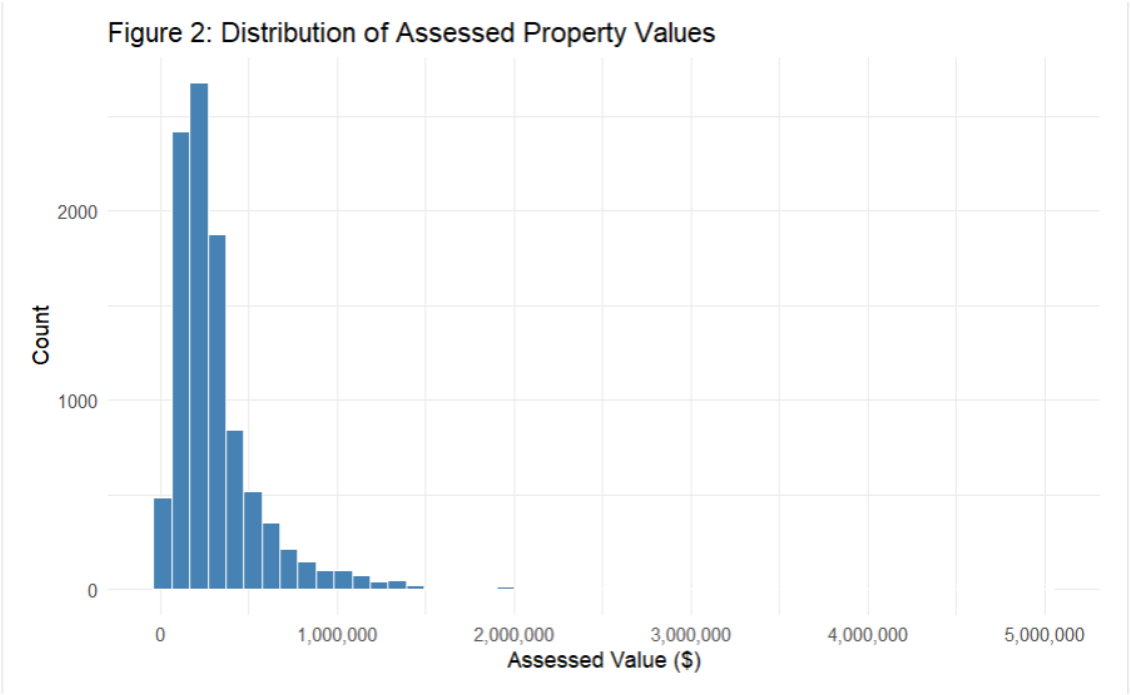

Figure 3: Actual vs Predicted Sale Price (OOB)

Scatter plot comparing actual and predicted prices. The red dashed line represents perfect prediction. Points cluster tightly around the line for lower-priced homes but diverge for luxury properties, indicating the model tends to underestimate high-value homes.